

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277968

Kind Code

A1

Publication Date

September 04, 2025

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MOVABLE DEVICE, IMAGE PROJECTION DEVICE, MOVING OBJECT, HEAD-MOUNTED DISPLAY, HEAD-UP DISPLAY, LASER HEADLAMP, OBJECT RECOGNITION DEVICE, AND OBJECT RECOGNITION APPARATUS, AND EYE TRACKING APPARATUS

Abstract

A movable device includes a movable portion, a driver to drive the movable portion, a first connect beam extending from the movable portion in a direction intersecting the center line of the movable portion. Further, there is a second connect beam extending from the driver in a direction intersecting the centerline of the movable portion, and an intermediate connector extending in a direction along the centerline and connecting the first connect beam and the second connect beam.

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Family ID: 96881120

Appl. No.: 19/066532

Filed: February 28, 2025

Foreign Application Priority Data

JP	2024-029671	Feb. 29, 2024
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Publication Classification

Int. Cl.: G02B26/08 (20060101); G02B26/10 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2024-029671, filed on Feb. 29, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] This present disclosure relates to a mobile device, a projection device, a moving object, a head-mounted display, a head-up display, a laser headlamp, an object recognition device, and a position detection device.

Related Art

[0003] A movable device such as a micro electro mechanical systems (MEMS) device manufactured by micromachining silicon or glass is known.

[0004] For example, in the related art, the technology in this field has been proposed that includes a mirror structure comprising (A) a mirror body and a mirror with a light reflecting layer provided on the surface of the mirror body, (B) a plurality of pillars with the upper end fixed to the back surface of the mirror body, (C) a displacement portion with one end fixed to the lower end of the respective pillar, and (D) a support portion with the other end of the displacement portion fixed.

[0005] However, in a movable device such as proposed in the related art, the rigidity is high to prevent distortion of the reflective surface. Therefore, in the related art, the reaction force applied to the mirror was not damped, resulting in suppression of the displacement of the mirror in the centerline direction, causing a decrease in the oscillation angle.

SUMMARY

[0006] According to an embodiment of the present disclosure, a movable device includes a movable portion, a driver to drive the movable portion, and a first connect beam extending from the movable portion in a direction intersecting the center line of the movable portion. Further, there is a second connect beam extending from the driver in a direction intersecting the centerline of the movable portion, and an intermediate connector extending in a direction along the centerline and connecting the first connect beam and the second connect beam.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0008] FIG. 1 is a plan view of a movable device according to a first embodiment of the present disclosure;

[0009] FIG. 2A is a cross-sectional view along the second axis L12 in FIG. 1;

[0010] FIG. 2B is a cross-sectional view along the second axis L11 in FIG. 1;

[0011] FIG. 3 is a cross-sectional view of a support portion, a driver, and a part of the connector, and the cross-sectional view along the second axis L12 in FIG. 1;

[0012] FIG. 4 is a cross-sectional view of a mirror and the part of the connector;

[0013] FIG. 5A is a schematic diagram of the movable device for a comparative example,

illustrating the state of the movable device when not driven;
[0014] FIG. 5B is a schematic diagram of the movable device for a comparative example, illustrating the state of the movable device when driven;
[0015] FIG. 6 is a schematic diagram of the mirror, the connector, and the driver, illustrating the state in which the reflective surface is inclined;
[0016] FIG. 7 is a plan view of the movable device according to a second embodiment of the present disclosure;
[0017] FIG. 8 is a cross-sectional view of the movable device according to a third embodiment of the present disclosure;
[0018] FIG. 9 is a cross-sectional view of the movable device according to a fourth embodiment of the present disclosure, illustrating a cut plane along the second axis L12;
[0019] FIG. 10 is a plan view of the mirror, the connector, and a part of the driver;
[0020] FIG. 11 is a cross-sectional view of the part of the connector and the driver;
[0021] FIG. 12 is a cross-sectional view of the movable device according to the fifth embodiment of the present disclosure;
[0022] FIG. 13 is a schematic diagram of the movable device according to the sixth embodiment of the present disclosure;
[0023] FIG. 14 is a schematic diagram illustrating an example of a light scan system;
[0024] FIG. 15 is a diagram illustrating a hardware configuration of a light scan system;
[0025] FIG. 16 is a functional block diagram illustrating an example of a control device;
[0026] FIG. 17 is a flowchart of an example of a process related to a light scan system;
[0027] FIG. 18 is a schematic diagram illustrating an example of an automobile including a head-up display apparatus;
[0028] FIG. 19 is a schematic diagram illustrating an example of a head-up display apparatus;
[0029] FIG. 20 is a schematic diagram illustrating an example of an image forming apparatus including an optical writing device;
[0030] FIG. 21 is a schematic diagram illustrating an example of an optical writing device;
[0031] FIG. 22 is a schematic diagram illustrating an example of an automobile including a laser imaging detection and ranging apparatus;
[0032] FIG. 23 is a schematic diagram illustrating an example of an automobile including a laser imaging detection and ranging apparatus;
[0033] FIG. 24 is a schematic diagram illustrating an example of a laser imaging detection and ranging apparatus;
[0034] FIG. 25 is a schematic diagram illustrating an example of a configuration of a laser headlamp;
[0035] FIG. 26 is a schematic perspective view of an example of a configuration of a head-mounted display;
[0036] FIG. 27 is a schematic diagram illustrating an example of a portion of a configuration of a head-mounted display;
[0037] FIG. 28 is a schematic diagram illustrating an example of a pupil-or-cornea position detection apparatus; and
[0038] FIG. 29 is a schematic diagram illustrating an example of a pupil-or-cornea position detection apparatus mounted on a head-mounted display.
[0039] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0040] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the

specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0041] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0042] The following is a description of the embodiment of the invention with reference to the drawings. In each drawing, the same reference signs are attached to the same configuration parts, and redundant description is appropriately simplified or omitted.

[0043] In the description of the following embodiments, pivoting, oscillating, and movable are to be considered synonymous. In each drawings, the mutually orthogonal X-axis, Y-axis, and Z-axis directions may be illustrated. The Z-axis direction is along the stacking direction of each layer in the piezoelectric driving portion. A view in the Z-axis direction may be described as a “plan view. In each drawing, a parallel oblique line may be added to a portion that is not a cross-section.

[0044] The X-axis direction includes the direction indicated by the arrow and vice versa; the direction in the X-axis direction to which the arrow points may be described as the +X direction and the direction opposite to the +X direction as the −X direction. The Y-axis direction includes the direction indicated by the arrow and vice versa; the direction in the Y-axis direction to which the arrow points may be described as the +Y direction and the direction opposite to the +Y direction as the −Y direction. The Z-axis direction includes the direction indicated by the arrow and vice versa; the direction in the Z-axis direction to which the arrow points may be described as the +Z direction and the direction opposite to the +Z direction as the −Z direction.

[0045] The terms “upper” and “lower” may also be used. The term “upper” may be in the +Z direction, and “lower” may be in the −Z direction. These directions do not limit the orientation of the movable device **13**, and the orientation of the movable device **13** is arbitrary. The movable device may be called a “light deflector”.

[0046] The movable device **13** according to the first embodiment will be described with reference to FIGS. **1** to **3**. FIG. **1** is a plan view of a movable device according to a first embodiment of the present disclosure. FIG. **2A** is a cross-sectional view along the second axis **L12** in FIG. **1**. FIG. **2B** is a cross-sectional view along the second axis **L11** in FIG. **1**. FIG. **3** is a cross-sectional view of a support portion, a driver, and a part of the connector, and the cross-sectional view along the second axis **L12** in FIG. **1**.

[0047] As illustrated in FIG. **1**, the movable device **13** has a mirror **101**, drivers **110A**, **110B**, **110C**, and **110D**, connectors **120A**, **120B**, **120C**, and **120D**, and a support portion **140**.

[0048] The mirror **101** has a reflective surface **14** that reflects incident light. The mirror **101** is an example of a movable portion. The mirror **101** has, for example, a disk-shaped mirror base (movable base) **102** and the reflective surface **14** formed on the mirror base **102**. The reflective surface **14** is formed on the +Z side of the mirror base **102**. The mirror base **102** forms, for example, a disk shape.

[0049] The support portion **140** forms a rectangular frame, viewed in the Z-axis direction. The mirror **101** is located at the center of the support portion **140** as viewed in the Z-axis direction. The support portion **140** supports the mirror **101** via a plurality of drivers **110A**, **110B**, **110C**, and **110D** and connectors **120A**, **120B**, **120C**, and **120D**.

[0050] The movable device **13** includes an electrode connector **150** that is electrically connected to the drivers **110A**, **110B**, **110C**, and **110D**. The electrode connector **150** has a plurality of pads.

[0051] The drivers **110A**, **110B**, **110C**, and **110D** are supported by the support portion **140**. The drivers **110A**, **110B**, **110C**, and **110D** extend inward from each side of the rectangular support portion **140** as viewed in the Z-axis direction. Here, the side closer to the mirror **101** is the inner side, and the side farther from the mirror **101** is the outer side.

[0052] The drivers **110A** and **110B** are arranged along the X-axis direction. The driver **110C** and

110D are arranged along the Y-axis direction. The mirror **101** is disposed between the drivers **110A** and **110B** in the X-axis direction. The mirror **101** is disposed between the drivers **110C** and **110D** in the Y-axis direction.

[0053] The drivers **110A**, **110B**, **110C**, and **110D** are connected to the mirror **101** via the corresponding connectors **120A**, **120B**, **120C**, and **120D**, respectively. The drivers **110A** and **110B** oscillate the mirror **101** around the first axis **L11** parallel to the Y-axis. The drivers **110C** and **110D** oscillate the mirror **101** around a second axis **L12** parallel to the X-axis.

[0054] As illustrated in FIG. 2A, the driver **110A** includes a piezoelectric element **111A** and a base **112A**. The piezoelectric element **111A** is disposed on an upper surface **112a** of the base **112A**. The driver **110B** includes a piezoelectric element **111B** and a base **112B**. The piezoelectric element **111B** is disposed on the upper surface **112a** of the base **112B**. As illustrated in FIG. 2B, the driver **110C** includes a piezoelectric element **111C** and a base **112C**. The piezoelectric element **111C** is disposed on the upper surface **112a** of the base **112C**. The driver **110D** includes a piezoelectric element **111D** and a base **112D**. The piezoelectric element **111D** is disposed on the upper surface **112a** of the base **112D**. The upper surface **112a** is an example of one of a pair of surfaces facing each other in the direction in which the centerline **O** of the base **112A** and **112B** extends. The lower surface **112b** is a surface opposite the upper surface **112a**. The drivers **110A**, **110B**, **110C**, and **110D** drive the mirror **101** (movable portion).

[0055] The connector **120A** connects the mirror **101** to the driver **110A**. the connector **120B** connects the mirror **101** and the driver **110B**. the connector **120C** connects the mirror **101** and the driver **110C**. and the connector **120D** connects the mirror **101** and the driver **110D**. The connector **120A** and **120B** are arranged along the X-axis direction. The connector **120C** and **120D** are arranged along the Y-axis direction.

[0056] As illustrated in FIG. 2A, the connector **120A** includes a first connect beam **121A**, a second connect beam **122A**, and an intermediate connector **123A**. The connector **120B** includes a first connect beam **121B**, a second connect beam **122B**, and an intermediate connector **123B**. As illustrated in FIG. 2B, the connector **120C** includes a first connect beam **121C**, a second connect beam **122C**, and an intermediate connector **123C**. The connector **120D** includes a first connect beam **121D**, a second connect beam **122D**, and an intermediate connector **123D**.

[0057] The first connect beams **121A** and **121B** extend from the mirror **101** in the X-axis direction. The first connect beams **121A** and **121B** extend in opposite directions. The first connect beams **121A** and **121B** are formed as an integral part of the mirror base **102**. The first connect beams **121A** and **121B** are formed to overhang from the mirror base **102**. The first connect beams of the connectors **120C** and **120D** include the same structure as the first connect beams **121A** and **121B**, and are not described here.

[0058] The second connect beam **122A** extends in the X-axis direction from the drive **110A** to the mirror **101**. The second connect beam **122A** is formed as an integral part of the base **112A**. The second connect beam **122A** is formed to overhang from the base **112A**.

[0059] The second connect beam **122B** extends out in the X-axis direction from the driver **110B** toward the mirror **101**. The second connect beam **122B** is formed as an integral part of the base **112B**. The second connect beam **122B** is formed to extend out from the base **112B**.

[0060] The second connect beam **122C** and **122D** include the same structure as the second connect beams **122A** and **122B**, so the description is omitted.

[0061] The intermediate connector **123A** extends in the Z-axis direction and connects the first connect beam **121A** to the second connect beam **122A**. The intermediate connector **123A** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The upper end of the intermediate connector **123A** may be connected to the end of the first connect beam **121A** that is farther from the mirror **101**. The lower end of the intermediate connector **123A** may be connected to the end of the second connect beam **122A** that is farther from the base **112A**. The intermediate connector **123A** extends from the second connect beam **122A** in the +Z direction.

[0062] The intermediate connector **123B** extends in the Z-axis direction and connects the first connect beam **121B** to the second connect beam **122B**. The intermediate connector **123B** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The upper end of the intermediate connector **123B** may be connected to the end of the first connect beam **121B** that is farther from the mirror **101**. The lower end of the intermediate connector **123A** may be connected to the end of the second connect beam **122B** that is farther from the base **112B**. The intermediate connector **123B** extends from the second connect beam **122B** in the +Z direction. As illustrated in FIG. 3, the length L_z and width W_x of the intermediate connector **123A** satisfy the relationship $L_z/W_x > 1$. In other words, the value of the length L_z of the intermediate connector **123A** is greater than the value of the width W_x ($L_z > W_x$). The width W_x can be the length of the intermediate connector **123A** in the x-axis direction or the diameter of the columnar intermediate connector **123A**.

[0063] For example, if the relationship $L_z/W_x = 1$ is satisfied, the rotation of the intermediate connector **123A** causes a decrease in the displacement of the mirror **101** in the Z-axis direction, which counteracts the displacement in the Z-axis direction generated by the driver **110A**. As a result, the oscillation angle θ of the mirror **101** decreases. An example of the oscillation angle θ is illustrated in FIGS. 5A and 5B which are explained in further detail below.

[0064] The intermediate connectors **120C** and **120D** have the same structure as the intermediate connectors **123A** and **123B**. The structures of intermediate connectors **120C** and **120D** are omitted.

[0065] The mirror **101**, drivers **110A**, **110B**, **110C**, and **110D**, connectors **120A**, **120B**, **120C**, and **120D**, and support portion **140** are formed from an SOI (Silicon On Insulator) substrate, for example, by applying an etching process, etc. to the SOI substrate. The mirror **101**, drivers **110A**, **110B**, **110C**, and **110D**, connectors **120A**, **120B**, **120C**, and **120D**, and support portion **140** are formed from the SOI substrate.

[0066] On the formed SOI substrate, a reflective surface **14**, and piezoelectric elements **111A**, **111B**, **111C**, and **111D**, and electrode connector **150** are formed. The reflective surface **14**, and piezoelectric elements **111A**, **111B**, **111C**, and **111D**, and electrode connector **150** are formed integrally with the SOI substrate. The reflective surface **14**, and the piezoelectric elements **111A**, **111B**, **111C**, and **111D**, and the electrode connector **150** may be formed after the SOI substrate is formed, or may be formed during the forming of the SOI substrate.

[0067] The SOI substrate includes a silicon support layer **161** made of single-crystal silicon (Si), a silicon oxide layer **162** formed on the silicon support layer **161** (+Z direction side), and a silicon active layer **163** made of single-crystal silicon formed on the silicon oxide layer **162**, as illustrated in FIGS. 3 and 4. The silicon oxide layer **162** can be referred to as a BOX (Buried Oxide) layer.

[0068] A member composed only the silicon active layer **163** functions as an elastic part having elasticity. The support portion **140** includes a silicon support layer **161**, a silicon oxide layer **162**, and a silicon active layer **163**. The bases **112A**, **112B**, and **112C**, and **112D** include a silicon active layer **163**.

[0069] The SOI substrate may not be planar and may have curvature. The components used to form the movable device **13** can be integrally formed by etching process and may be partially elastic substrates. The movable device **13** and are not limited to SOI substrates.

[0070] As illustrated in FIG. 4, the mirror base **102** includes, for example, a silicon active layer **163**. The reflective surface **14** includes a thin metal film including, for example, aluminum, gold, silver.

[0071] A movable thick-walled portion **103** for reinforcing the mirror portion is formed on the -Z side of the mirror portion base **102**. The movable thick portion **103** includes, for example, a silicon support layer **161** and a silicon oxide layer **162**. The movable thick portion **103** can suppress distortion of the reflective surface **14** caused by movement.

[0072] As illustrated in FIG. 3, the second connect beam **122A** includes a silicon active layer **163**. The second connect beam **122A** is formed as an integral part of the silicon active layer **163** of the

base **112A**. The second connect beam **122A** and the base **112A** may be formed from a single SOI substrate, for example.

[0073] As illustrated in FIG. 4, the first connect beams **121A** and **121B** include the silicon active layer **163**. The first connect beams **121A** and **121B** are formed as an integral part of the silicon active layer **163** of the mirror base **102**. The first connect beams **121A**, **121B** and the mirror base **102** may be formed from a single SOI substrate, for example.

[0074] The intermediate connectors **123A**, and **123B** include, for example, a silicon active layer **163**. The intermediate connectors **123A** and **123B** may be formed separately from the first connect beams **121A** and **121B** and the second connect beams **122A** and **122B**. The intermediate connector **123A** may be bonded to the first connect beam **121A** and the second connect beam **122A**, respectively. The intermediate connector **123B** may be bonded to the first connect beam **121B** and the second connect beam **122B**, respectively. The intermediate connectors **123A**, **123B** are bonded to the first connect beams **121A** and **121B** and the second connect beams **122A** and **122B** using an adhesive, such as a thermosetting resin or a light-curing resin.

[0075] The connectors **120A**, **12B**, **120C**, and **120D** can be fabricated using other specialized wafers or specialized processes. For example, two wafers can be joined to form a structure with two active layers.

[0076] As illustrated in FIG. 3, piezoelectric elements (piezoelectric portions) **111A**, **111B**, **111C**, and **111D** include a lower electrode **201**, a piezoelectric layer **202**, and an upper electrode **203**. The lower electrodes **201** are stacked on the bases **112A**, **112B**, **112C**, and **112D**. The piezoelectric layer **202** is stacked on the lower electrode **201**. The upper electrode **203** is stacked on the piezoelectric layer **202**. The piezoelectric layer **202** is sandwiched between the lower electrode **201** and the upper electrode **203** in the Z-axis direction. The lower electrode **201** and the upper electrode **203** include, for example, gold (Au) or platinum (Pt). The piezoelectric layer **202** includes, for example, a piezoelectric material, PZT (lead zirconate titanate).

[0077] The electrode connector **150** is formed on the surface of the support portion **140**. The electrode connector **150** is electrically connected to the lower electrode **201** and the upper electrode **203** of the piezoelectric elements **111A**, **111B**, **111C**, and **111D**. the electrode wiring connecting the lower electrode **201** and the upper electrode **203** to the electrode connector **150** is formed on the top surface of the support portion **140**. The electrode wiring is formed from aluminum (Al), for example. The lower electrode **201** and the upper electrode **203** are electrically connected to a control device **11** via the electrode wiring and the electrode connector **150**. The control device **11** applies a signal voltage to the lower electrode **201**. The upper electrode **203** is grounded (GND).

[0078] An insulating layer comprising a silicon oxide film may be formed on the +Z side of the upper electrode **203** and on at least one of the +Z side of the support portion **140**.

[0079] When electrode wiring is formed on the insulating layer, the insulating layer is not formed on the portion where the upper electrode **203** is connected to the electrode wiring and the portion where the lower electrode **201** is connected to the electrode wiring. Since the insulating layer is formed, the degree of freedom in designing the drivers **110A**, **110B**, **110C**, and **110D** and electrode wiring can be increased. Since the insulating layer is formed, short circuits caused by contact between electrodes can be suppressed. The silicon oxide film also functions as an antireflection material.

[0080] The movable device **13** may include piezoelectric elements for detection that detects elastic deformation of the drivers **110A**, **110B**, **110C**, and **110D** respectively.

[0081] The piezoelectric element **111A** deforms the base **112A** in response to an applied drive voltage. Similarly, the piezoelectric element **111B** deforms the base **112B** in response to the applied drive voltage. The piezoelectric element **111C** deforms the base **112C** in response to the applied drive voltage. The piezoelectric element **111D** deforms the base **112D** in response to the applied drive voltage.

[0082] The piezoelectric elements for detection output detection signals due to the piezoelectric

effect in response to the deformation of the bases **112A**, **112B**, **112C**, and **112D**. The piezoelectric elements for detection output the detection signals to the control device **11** via the electrode connector **150**.

[0083] FIG. 5A is a schematic diagram of the movable device for a comparative example, illustrating the state of the movable device when not driven. The movable device **1** for the comparative example has a mirror **101**, drive sections **110A** and **110B**, and connect beams **2A** and **2B**. The connect beam **2A** extends in the X-axis direction and connects the mirror **101** and the driver **110A**. The connect beam **2B** extends in the X-axis direction and connects the mirror **101** and the driver **110B**. the connect beams **2A** and **2B** do not include a first connect beam, a second connect beam, or an intermediate connector.

[0084] As illustrated in FIG. 5A, in the non-driven state, the centerline O along the direction orthogonal to the reflective surface **14** is along the Z-axis direction. As illustrated in FIG. 5B, in the driven state, the centerline O is inclined with respect to the Z-axis. In the movable device **1**, the drivers **110A** and **110B** can be driven to bend and deform the connect beams **2A** and **2B** into an arch shape, thereby displacing the swing angle θ of the mirror **101**.

[0085] The movable device **1** needs to improve the energy efficiency when oscillating the mirror **101**. In the mover **1**, if the drivers **110A** and **110B** are driven by applying a large voltage to the drivers **110A** and **110B**, the oscillation angle θ can naturally be increased. However, as the displacement of the drivers **110A** and **110B** increases, energy loss increases, and the movable device needs to improve energy efficiency.

[0086] In order to reduce the loss due to bending deformation in connect beams **2A** and **2B**, it is conceivable to fabricate connect beams **2A** and **2B** with soft materials. However, in the technical field of the movable device (optical deflector) **1** using micromachining technology, it is common to fabricate the entire movable device **1** with hard materials such as silicon.

[0087] FIG. 6 is a schematic diagram of the mirror, the connector, and the driver, illustrating the state in which the reflective surface is inclined. The connectors **120A** and **120B** of the movable device **13** include the first connect beams **121A** and **121B**, the second connect beams **122A** and **122B**, and the intermediate connectors **123A** and **123B**. In the movable device **1** of the comparative example of FIGS. 5A and 5B, the displacement of the mirror **101** to the swing angle was achieved by bending the connect beams **2A** and **2B**. In the movable device **13**, the rotation of the intermediate connectors **123A** and **123B** can replace part of the bending of the connect beams **2A** and **2B**. Therefore, the movable device **13** can reduce the bending of rigid members, an operation that requires a large amount of energy. This allows the movable device **13** to efficiently oscillate the mirror **101**. The movable device **13** can oscillate the mirror **101** with less energy than the movable device **1**.

[0088] The control by the control device **11** illustrated in FIG. 7 will be described. The control device **11** may have a controller **30** that is electrically connected to the electrode connector **150** and applies a drive voltage to the piezoelectric elements **111A**, **111B**, **111C**, and **111D** of the movable device **13**. As illustrated in FIG. 7, one of the pads of the electrode connector **150** is connected to the controller **30**, however all or part of the pads may be connected to the controller **30**.

[0089] The piezoelectric elements **111A**, **111B**, **111C**, and **111D** deform (e.g., expand or contract) in proportion to the potential of the applied voltage when a positive or negative voltage is applied in the polarization direction, and what is called reverse piezoelectric effect. The drivers **110** move the mirror **101** by utilizing the reverse piezoelectric effect.

[0090] In this case, the angle formed by the XY plane and the reflective surface **14** when the reflective surface **14** of the mirror **101** is inclined in the +Z or -Z direction relative to the XY plane is referred to as a deflection angle. A direction in which the +X-side of the mirror **101** is tilted in the +Z-direction is defined as a positive deflection angle, and a direction in which the +X-side of the mirror **101** is tilted in the -Z-direction is defined as a negative deflection angle.

[0091] In the drivers **110A** and **110B**, the respective piezoelectric layers **202** are deformed when

drive voltages are applied to the piezoelectric layers **202** in parallel via the lower electrodes **201** and upper electrodes **203**. The action of this deformation of the piezoelectric layers **202** causes the base **112A** and **112B** to flex and deform. As a result, a driving force around the first axis **L11** acts on the mirror **101** through the twisting of the two connectors **120A** and **120B**, and the mirror **101** oscillates around the first axis **L11**. The drive voltage applied to the drive sections **110A** and **110B** is controlled by the control device **11**.

[0092] The control device **11** can make the mirror **101** oscillates around the first axis **L11** with the period of the predetermined sinusoidal waveform drive voltage, by applying drive voltages of predetermined sinusoidal waveforms to the piezoelectric elements **111A** and **111B** in parallel.

[0093] The control device **11** and any control device or controller may be implemented using a processor, processing circuitry, or circuitry, which are described below.

[0094] For example, when the frequency of the drive voltage is set to approximately 20 kHz, which is the same as the resonant frequency of the connectors **120A** and **120B**, the mirror **101** can be made to resonate and vibrate at approximately 20 kHz by utilizing the mechanical resonance generated by the twisting of the connections **120A** and **120B**.

[0095] The movable device **13** of the first embodiment may be of a cantilever type in which the base **112A** and the connector **120A** extend from the support portion **140** toward the X-axis direction. The movable device **13** is not limited to the cantilever type. The movable device **13** need only be capable of oscillating the mirror **101** by the piezoelectric layer **202A** to which a drive voltage is applied. The movable device **13** may, for example, be of a double-ended support (double-held) type.

[0096] A movable device **13B** according to the second embodiment will be described. FIG. 7 is a plan view of the movable device according to a second embodiment of the present disclosure. The difference between the movable device **13B** according to the second embodiment and the movable device **13** according to the first embodiment is that an arrangement of the driving sections **110A**, **110B**, **110C**, and **110D** and the connectors **120A**, **120B**, **120C**, and **120D** is different. In the following description of the second embodiment, descriptions similar to those of the first embodiment are omitted.

[0097] The plurality of drivers **110A**, **110B**, **110C**, and **110D** are arranged in positions corresponding to the corners of the rectangular support portion **140**. In FIG. 7, the third axis **L13** and the fourth axis **L14** are illustrated, which intersect the first axis **L11** and the second axis **L12** at an angle of 45 degrees. The third axis **L13** and the fourth axis **L14** are orthogonal to each other. The third axis **L13** and fourth axis **L14** may be diagonals of the rectangular support portion **140**.

[0098] The driver **110A** and the driver **110B** face each other in the direction in which the third axis **L13** extends. The driving portion **110C** and driving portion **110D** face each other in the direction in which the fourth axis **L14** extends. The connectors **120A** and **120B** extend out from the mirror **101** in the direction in which third axis **L13** extends. The connectors **120C** and **120D** extend out from the mirror **101** in the direction in which the fourth axis **L14** extends.

[0099] The movable device **13B** according to the second embodiment has the same effect as the movable device **13** according to the first embodiment. The plurality of drive sections **110A**, **110B**, **110C**, and **110D**, and connectors **120A**, **120B**, **120C**, and **120D** may be arranged along the third axis **L13** or the fourth axis **L14**.

[0100] A movable device **13C** according to the third embodiment will be described. FIG. 8 is a cross-sectional view of the movable device according to a third embodiment of the present disclosure. The difference between the movable device **13C** according to the third embodiment and the movable device **13** according to the first embodiment is that an arrangement of the mirror **101** and the intermediate connectors **123E** and **123F** is different. In the following description of the third embodiment, descriptions similar to those of the first embodiment are omitted.

[0101] The mirror **101** of the movable device **13C** is disposed inside the support section **140** in the Z-axis direction. The support portion **140** has a predetermined length in the Z-axis direction. For

example, viewed in the Y-axis direction, the mirror **101** is disposed overlapping the support section **140**. The mirror **101** is disposed opposite the piezoelectric elements **111A** and **111B** to the bases **112A** and **112B** in the Z-axis direction. The mirror **101** is disposed in the $-Z$ direction from the lower surface **112B** of the bases **112A** and **112B**. The mirror **101** does not extend in the $+Z$ direction above the upper surface **112a** of the bases **112A** and **112B**. The reflective surface **14** is formed on the surface **102b** of the mirror base **102** in the $-Z$ direction. The reflective surface **14** is formed on the surface **102b** opposite the intermediate connectors **123E** and **123F** to the mirror base **102** in the $-Z$ direction.

[0102] The movable device **13C** includes the connectors **120E** and **120F**. The connector **120E** connects the mirror **101** and the driver **110A**. the connector **120F** connects the mirror **101** and the driver **110B**. Similarly, the movable device **13C** includes connectors connected to the drivers **110C** and **120D**, respectively, which have similar structure to the connectors **120E** and **120F**, so the description is omitted.

[0103] The connector **120E** includes a first connect beam **121A**, a second connect beam **122A**, and an intermediate connector **123E**. The connector **120F** has a first connect beam **121B**, a second connect beam **122B**, and an intermediate connector **123B**.

[0104] The intermediate connector **123E** extends in the Z-axis direction and connects the first connect beam **121A** to the second connect beam **122A**. The intermediate connector **123E** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The lower end of the intermediate connector **123E** may be connected to the end of the first connect beam **121A** that is farther from the mirror **101**. The upper end of the intermediate connector **123E** may be connected to the end of the second connect beam **122A** that is farther from the base **112A**. The intermediate connector **123E** extends from the second connect beam **122A** in the $-Z$ direction.

[0105] The intermediate connector **123F** extends in the $-Z$ direction and connects the first connect beam **121B** to the second connect beam **122B**. The intermediate connection **123F** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The lower end of the intermediate connector **123F** may be connected to the end of the first connect beam **121B** that is farther from the mirror **101**. The upper end of the intermediate connector **123F** may be connected to the end of the second connect beam **122B** that is farther from the base **112B**. The intermediate connector **123F** extends from the second connect beam **122B** in the $-Z$ direction.

[0106] The movable device **13C** according to the third embodiment also has the same effect as the movable device **13** according to the first embodiment. The mirror **101** may be disposed in the $-Z$ direction with respect to the drivers **110A** and **110B**. The intermediate connectors **123E** and **123F** may extend from the second connect beams **122A** and **122B** in the $-Z$ direction. In the movable device **13B** according to the second embodiment, the mirror **101** may be arranged in the $-Z$ direction with respect to the drive sections **110A** and **110B**.

[0107] A movable device **13D** according to the fourth embodiment will be described. FIG. **9** is a cross-sectional view of the movable device according to a fourth embodiment of the present disclosure, illustrating a cut plane along the second axis **L12** illustrated in FIGS. **10** and **11**. FIG. **10** is a plan view of the mirror, the connector, and a part of the driver. FIG. **11** is a cross-sectional view of the part of the connector and the driver. The difference between the movable device **13D** according to the fourth embodiment and the movable device **13** according to the first embodiment is that the movable device **13D** includes connectors **120G** and **120H** instead of the connectors **120A** and **120B**. In the following description of the fourth embodiment, explanations similar to those of the first embodiment are omitted.

[0108] The movable device **13D** includes the connectors **120G** and **120H**, as illustrated in FIG. **9**. The connector **120G** connects the mirror **101** to the driver **110A**. The connector **120H** connects mirror **101** to the driver **110B**. Similarly, the movable device **13D** includes connectors connected to the drivers **110C** and **110D**. The connectors connected to the drivers **110C** and **110D**, respectively, which have similar structure to the connectors **120G** and **120H**, so the description is omitted.

[0109] The connector **120G** includes a first connect beam **121G**, a second connect beam **122G**, and an intermediate connector **123G**. The connector **120H** includes a first connect beam **121H**, a second connect beam **122H**, and an intermediate connector **123H**.

[0110] The first connect beam **121G** has a pair of first beams **131**, a pair of first support beams **132**, and a first support panel **133**, as illustrated in FIG. **10**. The pair of first beams **131** extend in the X-axis direction and are separated in the Y-axis direction. The end of the first beams **131** closer to the mirror **101** is connected to the mirror **101**.

[0111] The pair of first support beams **132** extend inwardly from each other from the pair of first beams **131**. The pair of first support beams **132** extend in the Y-axis direction so as to be close to each other. The pair of first support beams **132** are connected to the end of the pair of first beams **131** that is farther from the mirror **101**. The width of the first support beams **132** intersecting in the longitudinal direction is narrower than the width of the first beams **131** intersecting in the longitudinal direction.

[0112] The first support panel **133** connects with the pair of first support beams **132**. The first support panel **133** is disposed between the pair of first support beams **132** in the Y-axis direction. The first support panel **133** has a disk shape. The intermediate connector **123G** is connected to the first support panel **133**.

[0113] The second connect beam **122G** includes a pair of second beams **134**, a pair of second support beams **135**, and a second support panel **136**, as illustrated in FIG. **11**. The pair of second beams **134** extend in the X-axis direction and are separated in the Y-axis direction. The end of the second beams **134** closer to the driver **110A** is connected to the driver **110A**.

[0114] As illustrated in FIG. **11**, the pair of second support beams **135** extend inwardly from each other from the pair of second beams **134**. The pair of second support beams **135** extend in the Y-axis direction so as to be close to each other. The pair of second support beams **135** are connected to the end of the pair of second beams **134** that is farther from the driver **110A**. The width of the second support beams **135** intersecting in the longitudinal direction is narrower than the width of the second beams **134** intersecting in the longitudinal direction.

[0115] The second support panel **136** connects with the pair of second support beams **135**. The second support panel **136** is disposed between the pair of second support beams **135** in the Y-axis direction. The second support panel **136** has a disk shape. The intermediate connector **123G** is connected to the second support plate **136**.

[0116] The intermediate connector **123G** extends in the Z-axis direction and connects the first connect beam **121G** and the second connect beam **122G**. The intermediate connector **123G** is, for example, columnar. The columnar shape may be a cylindrical or prismatic. The upper end of the intermediate connector **123G** may be connected to the first support panel **133** of the first connect beam **121G**. The lower end of the intermediate connector **123G** may be connected to the second support panel **136** of the second connect beam **122G**. The intermediate connection **123G** extends from the second support panel **136** of the second connect beam **122G** in the +Z direction.

[0117] The first connect beam **121H**, the second connect beam **122H**, and the intermediate connector **123H** of the connector **120H** have the same structure as the first connect beam **121G**, the second connect beam **122G**, and the intermediate connector **123G** of the connector **120G**, so the description is omitted.

[0118] The movable device **13D** according to fourth embodiment also has the same effect as the movable device **13** according to the first embodiment. The first connect beam **121G** may be configured with the pair of first beams **131** disposed apart in the Y-axis direction. The second connect beam **122G** may be configured with the pair of second beams **134** disposed apart in the Y-axis direction. The movable device **13B** according to the second embodiment and the movable device **13C** according to the third embodiment may be configured with the pair of first beams **131** and the pair of second beams **134**.

[0119] A movable device **13E** according to the fifth embodiment will be described. FIG. **12** is a

cross-sectional view of the movable device according to the fifth embodiment of the present disclosure. The difference between the movable device **13E** according to the fifth embodiment and the movable device **13** according to the first embodiment is that the movable device **13E** includes connectors **120I** and **120J** instead of connectors **120A** and **120B**. In the following description of the fifth embodiment, explanations similar to those of the first embodiment are omitted.

[0120] The movable device **13E** includes the connectors **120I** and **120J**, as illustrated in FIG. **12**. The connector **120I** connects the mirror **101** to the driver **110A**. The connector **120J** connects the mirror **101** to the driver **110B**. Similarly, the movable device **13E** includes connectors connected to the drivers **110C** and **110D**. The connectors connected to the drivers **110C** and **110D**, respectively, which have similar structure to the connectors **120I** and **120J**, so the description is omitted.

[0121] The connector **120I** includes the first connect beam **121A**, the second connect beam **122A**, the intermediate connector (first intermediate connector) **123A**, a third connect beam **124A**, and an intermediate connector (second intermediate connector) **125A**. The connector **120J** includes the first connect beam **121B**, the second connect beam **122B**, the intermediate connector (first intermediate connector) **123B**, a third connect beam **124B**, and an intermediate connector (second intermediate connector) **125B**. Similarly, connectors **120C** and **120D** include the first connect beam, the second connect beam, the intermediate connector (first intermediate connector), a third connect beam, and an intermediate connector (second intermediate connector), respectively.

[0122] The intermediate connector **123A** extends in the Z-axis direction and connects the first connect beam **121A** and the third connect beam **124A**. The intermediate connector **123A** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The upper end of the intermediate connector **123A** may be connected to the end of the first connect beam **121A** that is farther from the mirror **101**. The lower end of the intermediate connector **123A** may be connected to the third connect beam **124A**. The intermediate connector **123A** extends from the first connect beam **121A** in the -Z direction.

[0123] The third connect beam **124A** extends from the lower end of the intermediate connector **123A** in the X-axis direction. The third connect beam **124A** extends in the X-axis direction so as to approach the drive **110A**. The third connect beam **124A** connects the intermediate connector **123A** to the intermediate connector **125A**.

[0124] The intermediate connector **125A** extends in the Z-axis direction and connects the second connect beam **122A** and the third connect beam **124A**. The intermediate connector **125A** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The upper end of the intermediate connector **125A** may be connected to the end of the third connect beam **124A** that is farther from the intermediate connector **123A**. The lower end of the intermediate connector **125A** may be connected to the end of the second connect beam **122A** that is farther from the driver **110A**. The intermediate connector **125A** extends from the second connect beam **122A** in the +Z direction.

[0125] The intermediate connector **123B**, the third connect beam **124B**, and the intermediate connector **125B** of the connector **120J** have the same structure as the intermediate connector **123A**, the third connect beam **124A**, and the intermediate connector **125A** of the connector **120I**, so the description is omitted.

[0126] The movable device **13E** according to the fifth embodiment also has the same effect as the movable device **13** according to the first embodiment. The movable device **13E** may be configured with the third connect beams **124A** and **124B**, and the intermediate connectors **125A** and **125B**. The movable device **13B**, **13C**, and **13D** may also be configured with the third connect beams **124A** and **124B**, and the intermediate connectors **125A** and **125B**.

[0127] A movable device **13F** according to the sixth embodiment will be described. FIG. **13** is a schematic diagram of the movable device according to the sixth embodiment of the present disclosure. The difference between the movable device **13F** according to the sixth embodiment and the movable device **13** according to the first embodiment is that in the movable device **13F**, on the rear side of the mirror **101** includes a rear-side overhang portion **104** and a rear-side connector **105**.

In the following description of the sixth embodiment, descriptions similar to those of the first embodiment are omitted.

[0128] The mirror **101** includes the mirror base (movable base) **102**, the rear-side overhang portion **104**, and a rear-side connector **105**.

[0129] The rear-side overhang portion **104** extends from the back surface **102b**, which is the opposite side of the reflective surface **14**, in a direction along the center line O. The rear-side overhang portion **104** is, for example, columnar. The columnar shape may be cylindrical or prismatic. The rear-side overhang portion **104** may be other than columnar. The diameter of the rear-side overhang portion **104** may be smaller than the diameter of the reflective surface **14**, for example.

[0130] The rear-side connector **105** is connected to the rear-side overhang portion **104**. The rear-side connector **105** is, for example, disk-shaped. The rear-side connector **105** may be other shapes other than a disk shape. The rear-side connector **105** connects to first connect beams **121A** and **121B**. The first connect beam **121A** and **121B** may be connected to the rear-side overhang portion **104** instead of the rear-side connector **105**. The diameter of the rear-side connector **105** may be smaller than the diameter of the reflective surface **14**, for example.

[0131] The movable device **13F** according to the sixth embodiment also has the same effect as the movable device **13** according to the first embodiment.

[0132] In addition, in the movable device **13F**, the rear-side overhang portion **104** makes the reflective surface **14** project in the +Z direction. The projection of the reflective surface **14** can prevent the mirror **101** from contacting the connectors **120A** and **120B**, and the drivers **110A** and **110B** when the mirror **101** is oscillating.

[0133] In addition, in the movable device **13F**, the diameter of the reflective surface **14** is larger than the diameter of the rear-side connector **105**, which allows the reflective surface **14** to oscillate more efficiently. This is based on the fact that when the displacement **Z1** in the Z-axis direction generated by the drivers **110A** and **110B** is the same, the oscillation angle θ is larger when the diameter **R1** of the movable portion (rear-side connector **105**) to which the connectors **120A** and **120B** are directly connected is smaller than when the diameter **R1** is larger (see formula (1) below) ($\theta = \sin^{-1}$)

$$(\theta = \sin^{-1}(Z1/R1)) \quad (1)$$

[0134] In the movable device **13F**, the rear-side overhang portion **104** and the rear-side connector **105** may be formed separately or integrally. The movable device **13F** may be provided with the rear-side overhang portion **104** and without the rear-side connector **105**. The connectors **120A** and **120B** may be connected to the rear-side overhang portion **104**.

[0135] Embodiments of various applications to which the movable device **13**, **13B**, **13C**, **13D**, **13E**, or **13F** according to embodiments of the present disclosure is applied will be described below.

[0136] First, a light scan system **10** to which the movable device **13** according to an embodiment of the present disclosure is applied will be described in detail with reference to FIGS. **14** to **17**. FIG. **14** is a schematic diagram illustrating an example of the light scan system **10**. As illustrated in FIG. **14**, the light scan system **10** deflects the light beam emitted from a light source device **12** by the mirror surface **14** of the movable device **13** under the control of a control device **11**, and optically scans a scan surface **15** with the light beam.

[0137] The light scan system **10** includes the control device **11**, the light source device **12**, and the movable device **13** having the mirror surface **14**.

[0138] The control device **11** is an electronic circuit unit including, for example, a central processing unit (CPU) and a field-programmable gate array (FPGA). The movable device **13** is, for example, a micro electromechanical systems (MEMS) device having a mirror surface **14** and can move the mirror surface **14**. The light source device **12** is, for example, a laser device to emit a laser beam. The scan surface **15** is, for example, a screen.

[0139] The control device **11** generates a control command for the light source device **12** and the movable device **13** based on the obtained optical scan information, and outputs a drive signal to the light source device **12** and the movable device **13** based on the control command.

[0140] The light source device **12** irradiates an object with a light beam based on the input drive signal. The movable device **13** moves the mirror surface **14** in at least one of one-axial direction or two-axial directions based on the input drive signal.

[0141] Accordingly, for example, the mirror surface **14** of the movable device **13** is reciprocated in two-axial directions within a predetermined range by the control device **11** based on image information as an example of optical scan information so that the irradiation light beam from the light source device **12** that goes in the mirror surface **14** is deflected around a certain axis to perform optical scan. As a result, any image is projected onto the scan surface **15**.

[0142] A hardware configuration of an example of the light scan system **10** will be described with reference to FIG. **15**. FIG. **15** is a diagram illustrating a hardware configuration of a light scan system **10**. As illustrated in FIG. **15**, the light scan system **10** includes a control device **11**, a light source device **12**, and a movable device **13**, which are electrically connected. The control device **11** includes a CPU **20**, a random-access memory (RAM) **21**, a read-only memory (ROM) **22**, an FPGA **23**, an external input and output (I/F) **24**, a light source driver **25**, and a movable device driver **26**.

[0143] The CPU **20** is a computing device that reads programs and data from a storage device such as the ROM **22** to the RAM **21** and executes processing to implement the overall control and functions of the control device **11**. The RAM **21** is a volatile storage device to temporarily hold programs and data. The ROM **22** is a nonvolatile storage device that can retain programs and data even when the power is turned off, and stores processing programs and data executed by the CPU **20** to control the functions of the light scan system **10**. The FPGA **23** is a circuit that outputs control signals suitable for the light source driver **25** and the movable device driver **26** in accordance with the processing of the CPU **20**.

[0144] The external I/F **24** is, for example, an interface with an external device or a network. Examples of the external device include a host device such as a personal computer (PC), and a storage device such as a universal serial bus (USB) memory, an SD card, a compact disk (CD), a digital versatile disk (DVD), a hard disk drive (HDD), or a solid-state drive (SSD). The network is, for example, a controller area network (CAN) for an automobile, a local area network (LAN), and the Internet. The external I/F **24** may enable connection or communication with the external device, and the external I/F **24** may be prepared for each external device.

[0145] The light source driver is an electric circuit that outputs a drive signal such as a drive voltage to the light source device **12** in accordance with an input control signal. The movable device driver **26** is an electric circuit that outputs a drive signal such as a drive voltage to the movable device **13** in accordance with an input control signal. In the control device **11**, the CPU **20** acquires optical scan information from an external device or a network via the external I/F **24**. Any configuration is acceptable as long as the CPU **20** can acquire the optical scan information, and the optical scan information may be stored in the ROM **22** or the FPGA **23** in the control device **11**. Alternatively, for example, a storage device such as an SSD may be additionally disposed in the control device **11** and the optical scan information may be stored in the storage device.

[0146] The optical scan information is information indicating how the scan surface **15** is optically scanned, and for example, when an image is displayed by optical scan, the optical scan information is image data. In addition, for example, when optical writing is performed by optical scan, the optical scan information is writing data indicating a writing order and a writing position. Further, for example, when an object is recognized by optical scan, the optical scan information includes irradiation data indicating the irradiation timing and an irradiation range with a light beam for object recognition.

[0147] The control device **11** can implement a functional configuration described below by an instruction of the CPU **20** and the hardware configuration illustrated in FIG. **15**.

[0148] The functional configuration of the control device **11** of the light scan system **10** will be described with reference to FIG. **16**. FIG. **16** is a functional block diagram illustrating an example of a control device of the light scan system. As illustrated in FIG. **16**, the control device **11** includes a control unit **30** and a drive signal output unit **31** as functions.

[0149] The control unit **30** is implemented by, for example, the CPU **20** or the FPGA **23**, obtains the optical scan information from an external device, converts the optical scan information into a control signal, and outputs the control signal to the drive signal output unit **31**. For example, the control unit **30** obtains image data as optical scan information from, for example, an external device, generates a control signal from the image data by predetermined processing, and outputs the control signal to the drive signal output unit **31**. The drive signal output unit **31** is implemented by the light source driver **25** and the movable device driver **26**, and outputs a drive signal to the light source device **12** or the movable device **13** based on the input control signal.

[0150] The drive signal is a signal for controlling the drive of the light source device **12** or the movable device **13**. For example, in the light source device **12**, the drive voltage is used to control the irradiation timing and the irradiation intensity of the light source. In addition, for example, in the movable device **13**, the drive voltage is a drive voltage for controlling the timing and the movable range of the mirror surface **14** of the movable device **13**.

[0151] The process in which the light scan system **10** optically scans the scan surface **15** with a light beam will be described with reference to FIG. **17**. FIG. **17** is a flowchart of an example of a process related to an optical scanner.

[0152] In step **S11**, the control unit **30** obtains optical scan information from an external device. In step **S12**, the control unit **30** generates a control signal from the obtained optical scan information and outputs the control signal to the drive signal output unit **31**. In step **S13**, the drive signal output unit **31** outputs a drive signal to the light source device **12** and the movable device **13** based on the input control signal.

[0153] In step **S14**, the light source device **12** performs light irradiation based on the input drive signal. The movable device **13** moves the mirror surface **14** based on the input drive signal. The light source device **12** and the movable device **13** are driven so that the light beam is deflected in any direction, and optically scan is performed.

[0154] In the light scan system **10** described above, a single control device **11** includes a device and a function of controlling the light source device **12** and the movable device **13**, but the device and the function may be disposed separately from the control device for the light source device and the control device for the movable device.

[0155] In the light scan system **10**, the functions of the control unit **30** of the light source device **12** and the movable device **13** and the function of the drive signal output unit **31** are disposed in one control device **11**, but these functions may be disposed separately. For example, a drive signal output device including the drive signal output unit **31** is disposed separately from the control device **11** including the control unit **30**. The light scan system **10** described above may include an optical deflection system to perform light deflection by the movable device **13** having the mirror surface **14**, and the single control device **11**.

[0156] The light scan system **10** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability. As a result, the light scan system **10** can scan a wide range with a light beam with high accuracy.

Image Projection Apparatus

[0157] An image projection apparatus including the movable device according to the present embodiment will be described in detail with reference to FIGS. **18-23**.

[0158] FIG. **18** is a schematic view of an automobile **400** including a head-up display device **500** as an example of an image projection apparatus. FIG. **19** is a schematic diagram illustrating an example of a head-up display device **500**.

[0159] The image projection apparatus is an apparatus that projects an image by optical scan, and

is, for example, a head-up display device.

[0160] As illustrated in FIG. 18, the head-up display device 500 is installed, for example, near a windshield (e.g., a front window 401) of an automobile 400. The projection light beam L emitted from the head-up display device 500 is reflected by the front window 401 and directed to an observer (e.g., a driver 402) that is a user. Accordingly, the driver 402 can visually recognize the image projected by the head-up display device 500 as a virtual image. A combiner may be disposed on an inner wall surface of the windshield, and the user may visually recognize a virtual image by the projected light beam reflected by the combiner.

[0161] As illustrated in FIG. 19, in the head-up display device 500, a red laser light source 501R emits a red laser beam, a green laser light source 501G emits a green laser beam, and a blue laser light source 501B emits a blue laser beam. These three emitted laser beams pass through an incident optical system, and are deflected by a movable device 13 including a mirror surface 14. The incident optical system includes a collimator lens 502, a collimator lens 503, and a collimator lens 504, which are disposed for the red laser beam, the green laser beam, and the blue laser beam, respectively, two dichroic mirrors 505 and 506, and a light amount adjusting unit 507. The deflected laser beams are projected onto a screen through a projection optical system including a free-form surface mirror 509, an intermediate screen 510, and a projection mirror 511. In the head-up display device 500, the red laser light source 501R, the green laser light source 501G, and the blue laser light source 501B, the collimator lens 502, the collimator lens 503, the collimator lens 504, the dichroic mirror 505, and the dichroic mirror 506 are formed as a light source unit 530 by an optical housing.

[0162] The head-up display device 500 described above projects an intermediate image displayed on the intermediate screen 510 on the front window 401 of the automobile 400 so that the driver 402 can be visually recognized the intermediate image as a virtual image.

[0163] The red laser beam emitted from the red laser light source 501R, the green laser beam emitted from the green laser light source 501G, and the blue laser beam emitted from the blue laser light source 501B are collimated by the collimator lens 502, the collimator lens 503, and the collimator lens 504, respectively, as substantially parallel laser beams, and are combined by the two dichroic mirrors 505 and 506. The amount of the combined laser beam is adjusted by the light amount adjusting unit 507, and the movable device 13 including the mirror surface 14 two-dimensionally scans the free-form surface mirror 509 with the laser beam. The projection light beam L with which the movable device 13 two-dimensionally scans the free-form surface mirror 509 is reflected by the free-form surface mirror 509 is corrected in distortion, and is condensed on the intermediate screen 510 to display an intermediate image. The intermediate screen 510 includes a microlens array in which microlenses are two-dimensionally arranged, and enlarges the projection light beam L incident on the intermediate screen 510 at each microlens.

[0164] The movable device 13 reciprocates the mirror surface 14 in two-axial directions and two-dimensionally scans the free-form surface mirror 509 with the projection light beam. The control of the movable device 13 is performed in synchronism with the timing of light emission of the red laser light source 501R, the green laser light source 501G, and the blue laser light source 501B.

[0165] As described above, the head-up display device 500 has been described as an example of the image projection apparatus, but the image projection apparatus may be any device that projects an image by performing optical scan with the movable device 13 including the mirror surface 14. For example, the image projection apparatus can be applied to a projector that is placed on a desk and projects an image on a display screen, and a head-mounted display device that projects an image to a reflective-and-transmissive screen included in a mounting member mounted on an observer's head or projects an image to an eyeball as a screen.

[0166] The image projection apparatus may be included in not only a vehicle and a mounting member but also, for example, a moving body such as an aircraft, a ship, and a mobile robot, or a non-moving body such as a work robot that operates a drive object such as a manipulator without

moving from the site.

[0167] The head-up display device **500** is an example of a head-up display. The automobile **400** is an example of a vehicle.

[0168] Since the head-up display device **500** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, a large screen image with high image quality can be displayed.

[0169] An optical writing apparatus to which the movable device **13** of the present embodiment is applied will be described in detail with reference to FIGS. **20** and **21**.

[0170] FIG. **21** is a schematic diagram illustrating an example of an image forming apparatus including an optical writing apparatus **600**. FIG. **21** is a schematic diagram illustrating an example of the optical writing apparatus **600**.

[0171] As illustrated in FIG. **21**, the optical writing apparatus **600** is used as a unit of an image forming apparatus such as a laser printer **650** having a printer function using a laser beam. In the image forming apparatus, the optical writing apparatus **600** optically scans a photoconductive drum that serves as a scan surface **15** with one or multiple laser beams, to perform optical writing to the photoconductive drum.

[0172] As illustrated in FIG. **21**, in the optical writing apparatus **600**, a laser beam from a light source device **12** such as a laser element passes through an imaging optical system **601** such as a collimator lens, and is deflected in one-axial direction or two-axial directions by a movable device **13** including a mirror surface **14**. The laser beam deflected by the movable device **13** passes through the scanning optical system **602** including a first lens **602a**, a second lens **602b**, and a reflecting mirror portion **602c**, and the optical writing apparatus **600** irradiates the scan surface **15** (e.g., photoconductive drum or photosensitive paper) with the laser beam to perform optical writing. The scanning optical system **602** forms an image of a light beam in a spot shape on the scan surface **15**. The light source device **12** and the movable device **13** including the mirror surface **14** are driven based on the control of the control device **11**.

[0173] As described above, the optical writing apparatus **600** can be used as a member of an image forming apparatus having a printer function by a laser beam. Further, the optical writing apparatus **600** can change the scanning optical system to perform optical scan not only in one-axial direction but also two-axial directions, and thus can be used as a unit of the image forming apparatus such as a laser label apparatus that deflects the laser beam to a thermal medium to print by heating.

[0174] Since the movable device **13** including the mirror surface **14** applied to the optical writing apparatus has a smaller power consumption for driving than a rotary polygon mirror using a polygon mirror, and the movable device **13** has an advantage in saving power of the optical writing apparatus. In addition, since the wind noise generated when the movable device **13** is vibrated is smaller than the wind noise of the rotary polygon mirror, it is advantageous for increasing the quietness of the optical writing apparatus. The optical writing apparatus requires a much smaller installation space than the rotary polygon mirror, and the movable device **13** generates a small amount of heat, so that the optical writing apparatus can be easily reduced in size. As a result, the image forming apparatus has an advantage in reducing in size.

[0175] Since the optical writing apparatus **600** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, an image with high image quality can be formed on a large region of a recording medium.

[0176] An object recognition apparatus to which the movable device of the present embodiment is applied will be described in detail with reference to FIGS. **22** and **23**.

[0177] FIG. **22** is a schematic diagram illustrating an automobile in which a laser imaging detection and ranging (LiDAR) apparatus that is an example of an object recognition apparatus is mounted. In the schematic diagram, the LiDAR apparatus is mounted on a body of the automobile.

[0178] FIG. **23** is a schematic diagram illustrating an automobile in which a laser imaging detection and ranging (LiDAR) apparatus that is an example of an object recognition apparatus is

mounted. In the schematic diagram, the LiDAR apparatus is mounted on a lighting unit including a headlight of the automobile.

[0179] FIG. **24** is a schematic diagram illustrating an example of a LiDAR apparatus. The object recognition apparatus is an apparatus that recognizes an object in a target direction, and is, for example, a LiDAR apparatus. As illustrated in FIGS. **22** and **23**, the LiDAR apparatus **700** is mounted on, for example, an automobile **701**, and scans a region with a light beam in a predetermined direction, and recognizes the object **702** in the region by receiving the reflected light beam from the object **702**.

[0180] As illustrated in FIG. **24**, the movable device **13** includes an incident optical system including a collimate lens **703** that is an optical system to form a divergent light beam into a substantially parallel light beam, and a plane mirror **704**. The movable device **13** including the mirror surface **14** scans an object with the laser beam that is emitted from the light source device **12** and passes through the incident optical system in a one-axial direction or two-axial directions. The substantially parallel light deflected by the movable device **13** reaches the object **702** in front of the LiDAR apparatus **700** via a projection lens **705** that is a projection optical system. The control device **11** drives the light source device **12** and the movable device **13**. The light detector **709** detects the reflected light beam reflected by the object **702**. In other words, the reflected light beam passes through a condenser lens **706** that is an optical system for receiving and detecting the incident light beam, and is received by an image sensor **707**. The image sensor **707** outputs a detected signal to a signal processing circuit **708**. The signal processing circuit **708** performs predetermined processing such as binarization or noise processing on the received detection signal, and outputs the result to a distance measurement circuit **710**.

[0181] The distance measurement circuit **710** determines the presence or absence of the object **702** according to a time difference between a timing at which the light source device **12** emits the laser beam and a timing at which the light detector **709** receives the laser beam, or a phase difference for each pixel of the image sensor **707** that receives the laser beam, and calculates distance information with respect to the object **702**.

[0182] Since the movable device **13** including the mirror surface **14** is less likely to be damaged as compared with a polygon mirror and is smaller in size, a smaller LiDAR apparatus having higher durability can be provided. When the LiDAR apparatus is attached to, for example, a vehicle, an aircraft, a ship, or a robot, the presence or absence of an obstacle or an object can be determined or the distance to the obstacle or the object can be recognized by performing optical scan within a predetermined range.

[0183] In the object recognition apparatus described above, the LiDAR apparatus **700** is described as an example. However, the object recognition apparatus is not limited to the embodiment described above as long as the object recognition apparatus performs optical scan by the movable device **13** including the mirror surface **14** under control of the control device **11** and receives the reflection light beam by the light detector to recognize the object **702**.

[0184] The object recognition apparatus described above can be applied to, for example, a biometric authentication apparatus that calculates object information such as a shape from the distance information obtained by scanning a hand or a face and recognizes the object by referring the record, a security sensor that recognizes an intruder by optically scanning the object region, or a unit of the three-dimensional scanner that outputs three-dimensional data.

[0185] Since the object recognition apparatus includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, the object recognition apparatus can recognize an object in a wide range with high accuracy.

[0186] A laser headlamp **50** in which the movable device of the present embodiment is applied to a headlight of an automobile will be described with reference to FIG. **25**. FIG. **25** is a schematic diagram illustrating an example of a configuration of the laser headlamp **50**.

[0187] The laser headlamp **50** includes a control device **11**, a light source device **12b**, a movable

device **13** including a mirror surface **14**, a mirror **51**, and a transparent plate **52**.

[0188] The light source device **12b** is a light source that emits a blue laser beam. The light beam emitted from the light source device **12b** goes in the movable device **13** and is reflected by the mirror surface **14**. The movable device **13** moves the mirror surface in the X- and Y-directions based on the signal from the control device **11**, and two-dimensionally scans an object with the blue laser beam from the light source device **12b** in the X- and Y-directions.

[0189] The scan light beam by the movable device **13** is reflected by the mirror **51** and goes in the transparent plate **52**. The front surface or the back surface of the transparent plate **52** is coated with a yellow fluorescent material. The blue laser beam from the mirror **51** changes to white in the legalized range of the color of the headlight when passing through the yellow phosphor coating of the transparent plate **52**. Thus, the area ahead of the automobile is illuminated with the white light beam from the transparent plate **52**.

[0190] The scan light beam by the movable device **13** is scattered in a predetermined manner when passing through the fluorescent material of the transparent plate **52**. Accordingly, the glare of the illumination object ahead of the automobile is reduced.

[0191] When the movable device **13** is applied to a headlight of an automobile, the colors of the light source device **12b** and the fluorescent material are not limited to blue and yellow, respectively. For example, the light source device **12b** may be a near-ultraviolet light source, and the transparent plate **52** may be coated with a mixture of uniformly mixed phosphors of blue, green, and red, which are the three primary colors of light. In this case, the light beam passing through the transparent plate **52** can be converted into white light, and the area ahead of the automobile can be illuminated with white light beam.

[0192] Since the laser headlamp **50** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, the laser headlamp can irradiate a wide region with a laser beam with high accuracy.

[0193] A head-mounted display (HMD) **60** to which the movable device of the present embodiment is applied will be described with reference to FIGS. **25** and **26**. The HMD **60** is a head-mounted display that can be mounted on the human head, and has a shape of, for example, glasses.

[0194] FIG. **26** is a perspective view of an external appearance of the HMD **60**. In FIG. **26**, the HMD **60** includes fronts **60a** symmetrically disposed in a pair on the left and right sides and temples **60b**. The fronts **60a** can be formed by, for example, light guide plates **61**, and an optical system and a control device can be built in the temples **60b**.

[0195] FIG. **27** is a diagram illustrating a partial configuration of the HMD **60**. Although FIG. **27** illustrates the partial configuration for the left eye, the HMD **60** has the same configuration for the right eye.

[0196] The HMD **60** includes a control device **11**, a light source unit **530**, a light amount adjusting unit **507**, a movable device **13** including a mirror surface **14**, a light guide plate **61**, and a half mirror **62**.

[0197] As described above, the light source unit **530** includes the red laser light source **501R**, the green laser light source **501G**, the blue laser light source **501B**, the collimator lens **502**, the collimator lens **503**, the collimator lens **504**, the dichroic mirror **505**, and the dichroic mirror **506** in an optical housing as a unit. In the light source unit **530**, the red laser beam from the red laser light source **501R**, the green laser beam from the green laser light source **501G**, and the blue laser beam from the blue laser light source **501B** are combined by the dichroic mirror **505** and the dichroic mirror **506**. The light source unit **530** emits the combined parallel light beam.

[0198] The light beam from the light source unit **530** is adjusted by the light amount adjusting unit **507** in its light amount and goes in the movable device **13**. The movable device **13** moves the mirror surface **14** in the X- and Y-directions based on the signal from the control device **11**, and two-dimensionally scans an object with the laser beam from the light source unit **530**. The control of the movable device **13** is performed in synchronism with the timing of light emission of the red

laser light source **501R**, the green laser light source **501G**, and the blue laser light source **501B**, and a color image is formed by optical scan.

[0199] The scan light beam by the movable device **13** goes in the light guide plate **61**. The light guide plate **61** guides the scan light beam to the half mirror **62** while reflecting the scan light beam on the inner wall surfaces. The light guide plate **61** is made of a resin having a transparency with respect to the wavelength of the scan light beam.

[0200] The half mirror **62** reflects the light beam from the light guide plate **61** to the back surface side of the HMD **60** and emits the light beam in the direction of the eye of a wearer **63** of the HMD **60**. The half mirror **62** has, for example, a free-form surface shape. The image by the scan light beam is reflected by the half mirror **62**, and is imaged on the retina of the wearer **63**. Alternatively, the image is formed on the retina of the wearer **63** by reflection at the half mirror **62** and the lens effect of the lens of the eyeball. The spatial distortion of the image is corrected by the reflection at the half mirror **62**. The wearer **63** can observe the image formed with the light beam scanned in the X- and Y-directions.

[0201] Since the half mirror **62** is used, the wearer **63** observes a superimposed image of an image of a light beam from the outside and an image by scan light beam. A mirror may be used instead of the half mirror **62** so that the light beam from the outside is eliminated, and only an image by the scan light beam can be observed.

[0202] Since the HMD **60** includes the movable device **13** that can obtain a large displacement amount while increasing the displacement stability, the HMD **60** can display a large screen image with high image quality.

[0203] A tilt-position detection apparatus of an eyeball that includes the movable device **13** will be below. The tilt-position detection apparatus for the eyeball is a pupil-or-cornea position detection apparatus **80**. FIG. **28** is a schematic block diagram illustrating an example of a pupil-or-cornea position detection apparatus **80**. FIG. **29** is a schematic block diagram illustrating an example of the pupil-or-cornea position detection apparatus **80** mounted on a head-mounted display.

[0204] The “tilt-position of the eyeball” in the present embodiment is the position of the pupil-or-cornea of the eyeball or the direction of the user's line of sight. In the following description, the “tilt-position of the eyeball” is the position of the pupil or the cornea, and the “tilt-position detection apparatus of the eyeball” is the “pupil-or-cornea position detection apparatus.” The pupil-or-cornea position detection apparatus described below is the same as a line-of-sight direction tracking apparatus (eye tracking apparatus) that detects or tracks the direction of the user's line of sight continuously or at intervals of time.

[0205] The pupil-or-cornea position detection apparatus **80** illustrated in FIG. **28** includes a light source **82**, a first light deflection part **83**, a movable device **13**, a second light deflection part **85**, and a light receiving unit **86**.

[0206] The light source **82** includes, for example, a red laser light source **82r** to emit a red laser beam, a green laser light source **82g** to emit a green laser beam, a blue laser light source **82b** to a blue laser light beam, and an infrared laser light source **82ir** to emit an infrared laser beam. The red laser light sources **82r**, the green laser light source **82g**, and the blue laser light source **82b** may be any one or a combination of two. The red laser light sources **82r**, the green laser light source **82g**, and the blue laser light source **82b** emit laser beams to form an image by the movable device **13**.

[0207] The infrared laser light source **82ir** emits a light beam to detect the position of the pupil or the cornea. The light beam for detecting the position of the pupil or the cornea is not limited to an infrared light beam, and may be a visible light beam. The light beam for detecting the position of the pupil or the cornea is preferably an invisible light beam from the viewpoint of increasing the visibility of a drawn image.

[0208] The first light deflection part **83** is, for example, a dichroic mirror, and deflects the light beam emitted from the light source **82** toward the mirror surface **14** of the movable device **13** while combining the light beams. The pupil-or-cornea position detection apparatus **80** may include

multiple first light deflection parts **83-1**, **83-2**, **83-3**, **83-4**, and **83-5**, in accordance with the number of the laser light sources (i.e., the red laser light sources **82r**, the green laser light source **82g**, the blue laser light source **82b**, and the infrared laser light source **82ir**). The first light deflection part **83** includes multiple first light deflection parts **83-1**, **83-2**, **83-3**, **83-4**, and **83-5**. The multiple first light deflection parts **83-1**, **83-2**, **83-3**, **83-4**, and **83-5** deflect the light beams while combining the light beams.

[0209] The movable device **13** includes a mirror surface **14** and scans the second light deflection part **85** with the light beam deflected by the first light deflection part **83** in two-dimensional directions. At this time, the movable device **13** scans the second light deflection part **85** with the light beam deflected by the first light deflection part **83** using, for example, raster scan to form an image. The movable device **13** can scan the second light deflection part **85** with the light beam deflected by the first light deflecting part **83** by spiral scan.

[0210] The second light deflection part **85** is, for example, a holographic optical element, and deflects the light beam L1 with which the movable device **13** scanned toward the eyeball **87** of the user. At least a portion of the light beam L2 deflected by the second light deflection part **85** goes in the eyeball **87** of the user as display image light beam. The second light deflection part **85** may include multiple light deflection members. For example, multiple light deflection parts that reflect specific light beams among the light beams emitted from the light source **82** may be used, and the mirror surface may be different for each light beams emitted from the light source **82**. As a specific example, a structure in which a light deflection part that reflects light beams emitted from the red laser light source **82r**, the green laser light source **82g**, and the blue laser light source **82b** and a light deflection member for reflecting the light beam emitted from the infrared laser light source **82ir** are laminated in order of proximity to the eyeball **87** can be used.

[0211] The light receiving unit **86** receives the light beam L3 reflected by the eyeball **87** of the user among the light beam L2 deflected by the second light deflection part **85**, and outputs a detection signal SD corresponding to the received light beam. The light receiving unit **86** is an image sensor that can detect, for example, an infrared light beam. The light receiving unit **86** may include multiple light receiving units and be disposed at positions that can receive the light beam L3 reflected by the eyeball **87** of the user. The light beam received by the light receiving unit **86** changes in intensity depending on the change in position of the eyeball (e.g., pupil or cornea), that is, the change in the direction of the line of sight. Thus, the pupil-or-cornea position detection apparatus **80** of the pupil or the cornea in the present embodiment detects or estimates the position of the pupil or the cornea based on the intensity of the light beam received by the light receiving unit **86**. The light receiving unit **86** may be configured to image the eyeball **87** irradiated with the light beam L2 deflected by the second light deflection part **85**. In this case, the pupil-or-cornea position detection apparatus **80** detects or estimates the tilt-position of the eyeball **87** based on the position of the pupil or cornea included in the captured image (i.e., detection signal SD) and the position at which the light beam L2 deflected by the second light deflection part **85** is reflected in the eyeball **87**.

[0212] As described above, the pupil-or-cornea position detection apparatus **80** according to the present embodiment can detect the position of the pupil or cornea while forming an image by the movable device **13**. Further, since the movable device **13** has a configuration that can perform optical scan more efficiently, image formation and detection of the position of the pupil or cornea can be implemented with lower power. Further, the movable device **13** can obtain the above-described effect without changing the area necessary for mounting the pupil-or-cornea position detection apparatus **80** as compared with the configuration in the related art. Accordingly, a configuration in which the size of the pupil-or-cornea position detection apparatus **80** is not increased can be implemented.

[0213] The pupil-or-cornea position detection apparatus **80** can be mounted on a head-mounted display as, for example, an eye tracking apparatus, and can detect or track the direction of the user's

line of sight. In this case, for example, the resolution of an image displayed in a region near the direction of the user's line of sight in other regions is reduced (i.e., foveal rendering) so that the image processing can be speeded up as compared with the case where a high-resolution image is displayed in the entire region.

[0214] The pupil-or-cornea position detection apparatus **80** illustrated in FIG. **29** includes a light source **82**, first light deflectors **83-1-4**, a lens **92**, a lens **93**, a scanning mirror **94**, a deflecting mirror **95**, a second light deflector **85**, a light receiver **86**, and a controller **96**. The light source **82**, the first optical deflectors **83-1-4**, the second light deflector **85**, and the light-receiver **86** have similar structure to the configuration described in FIG. **28**, so the description is omitted.

[0215] The lens **92** is an optical system that converts the light emitted by light source **82** into an abbreviated collimated light. The lens **93** is an optical system that forms the light converted by the lens **92** into the desired laser beam state. In this embodiment, the pupil-or-cornea position detection apparatus **80** includes both the lens **92** and the lens **93**. However, it is not necessary that the pupil-or-cornea position detection apparatus **80** include the lens **92** or the lens **93**.

[0216] The light formed by the lens **92** and the lens **93** enters the scanning mirror **94** (movable device **13**). The scanning mirror **94** scans the incident light to form image light. The formed image light is incident on the deflecting mirror **95** and reflected in the direction toward the second light deflector **85**. The deflecting mirror **95** corresponds to the first light deflector **83-5** illustrated in FIG. **28**, but it is preferable to have a structure that can run light with a movable device **13**. To make the deflecting mirror **95** a light-scannable configuration the image can be projected over a wider area.

[0217] Although, the deflecting mirror **95** illustrated in FIG. **29** is disposed between the scanning mirror **94** and the second optical deflector **85**, the arrangement of the deflecting mirror **95** and the scanning mirror **94** is not limited to this. For example, the scanning mirror **94** is disposed between the deflecting mirror **95** and the second light deflecting section **85**, and the light reflected by the deflecting mirror **95** is scanned in the 2-axis direction by the scanning mirror **94**. The controller **96** also controls the luminescence and light intensity of the light source **82** by giving the formation drive signal **SL1** to the light source **82** and drives the scanning mirror **94** by giving the scanning drive signal **SS** to the scanning mirror **94**. If the deflecting mirror **95** is configured for optical scanning, the controller **96** gives the deflection drive signal **ST** to the deflecting mirror **95** to drive the deflection mirror **95** in order to control the image projection position according to the acquired gaze information.

[0218] As described above, preferable embodiments have been described in detail. However, embodiments according to the present disclosure are not limited to the embodiments described above, and various modifications and substitutions can be made to the above-described embodiments of the present disclosure without departing from the scope of the claims.

[0219] All the numerals such as ordinal numbers and numbers and reference symbols used in the description of the embodiments are illustrative for specifically describing the technique of the present invention, and the present invention is not limited to the illustrated numerals. In addition, a coupling relation between the components is an example for specifically describing the technology of the present disclosure, and a connection relation for implementing a function of the present disclosure is not limited thereto.

[0220] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, ASICs ("Application Specific Integrated Circuits"), FPGAs ("Field-Programmable Gate Arrays"), and/or combinations thereof which are configured or programmed, using one or more programs stored in one or more memories, to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry

out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein which is programmed or configured to carry out the recited functionality.

Claims

1. A movable device comprising: a movable portion; a driver to drive the movable portion; a first connect beam extending from the movable portion in a direction intersecting a centerline of the movable portion; a second connect beam extending from the driver in a direction intersecting the centerline of the movable portion; and an intermediate connector extending in a direction along the centerline and connecting the first connect beam and the second connect beam.
2. The movable device according to claim 1, wherein: the intermediate connector is columnar; and the intermediate connector extends in a direction along the centerline.
3. The movable device according to claim 2, wherein: the driver includes: a piezoelectric portion; and a base connected to the second connect beam, the piezoelectric portion is disposed on one side of the base in the direction along the centerline, and the intermediate connector extends from the second connect beam to a side opposite to the piezoelectric portion.
4. The movable device according to claim 3, further comprising: a support which includes a frame to support a plurality of the drivers, wherein: the movable portion is disposed inside the support as viewed along the centerline, the base extends inwardly from the support as viewed along the centerline, the support has a predetermined length in the direction along the centerline, and the movable portion is disposed inside the support as viewed along the centerline.
5. The movable device according to claim 1, further comprising: a first support panel, wherein the first connect beam includes: a pair of beams extending in the longitudinal direction of the first connect beam, and disposed apart in the width direction that intersects the longitudinal direction of the first connect beam; and a pair of support beams extending inwardly from each other from the pair of beams, wherein: the first support panel connects with the pair of support beams, and the intermediate connector is connected to the first support panel.
6. The movable device according to claim 1, further comprising: a second support panel, wherein the second connect beam includes: a pair of beams extending in the longitudinal direction of the second connect beam, and disposed apart in the width direction that intersects the longitudinal direction of the second connect beam; and a pair of support beams extending inwardly from each other from the pair of beams, wherein: the second support panel connects with the pair of support beams, and the intermediate connector is connected to the second support panel.
7. The movable device according to claim 1, wherein the intermediate connector includes: a first intermediate connector; a second intermediate connector; and a third connect beam, wherein the first intermediate connector and the second intermediate connector are connected via a third connect beam.
8. The movable device according to claim 1, wherein the movable portion includes: a reflective surface; a movable base including the reflective surface; and a rear-side connector disposed on a rear side of the movable base, which is opposite side of the reflective surface, wherein: the rear-side connector is connected to the first connector, and a diameter of the reflective surface is larger than a diameter of the rear-side connector.
9. The movable device according to claim 8, wherein the movable portion includes: a rear-side overhang portion extending from the rear side of the movable base in a direction along the centerline, wherein the rear-side overhang portion is connected to the first connect beam via the rear-side connector.
10. An image projection apparatus comprising: a light scan system including the movable device according to claim 1.
11. A moving body comprising: the image projection apparatus according to claim 10.
12. A head-mounted display comprising: the movable device according to claim 1.

- 13.** A head-up display comprising: the movable device according to claim 1.
 - 14.** A laser headlamp comprising: the movable device according to claim 1.
 - 15.** An object recognition apparatus comprising: the movable device according to claim 1.
 - 16.** An eye tracking apparatus comprising: the movable device according to claim 1.
 - 17.** A device comprising: a mirror; a driver to drive the mirror; a first connect beam extending from the mirror in a direction intersecting a centerline of the mirror; a second connect beam extending from the driver in a direction intersecting the centerline of the mirror; and an intermediate connector extending in a direction along the centerline and connecting the first connect beam and the second connect beam.
 - 18.** The device according to claim 17, wherein: the intermediate connector is columnar; and the intermediate connector extends in a direction along the centerline.
 - 19.** The device according to claim 18, wherein: the driver includes: a piezoelectric portion; and a base connected to the second connect beam, the piezoelectric portion is disposed on one side of the base in the direction along the centerline, and the intermediate connector extends from the second connect beam to a side opposite to the piezoelectric portion.
 - 20.** The device according to claim 19, further comprising: a support which includes a frame to support a plurality of the drivers, wherein: the mirror is disposed inside the support as viewed along the centerline, the base extends inwardly from the support as viewed along the centerline, the support has a predetermined length in the direction along the centerline, and the movable portion is disposed inside the support as viewed along the centerline.
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